

More Efficient and Greener

Ever more advanced turbofan engines are being developed to power new generations of lighter, more efficient airliners, further fuel savings being made as the airlines move away from the “hub and spoke” system to direct flights using twin-engine jets. For short-range aircraft, the focus is now on developing alternative power systems.

▽ Cirrus Vision SF50 2016

Origin US

Engine 1,800 lb (817 kg) thrust Williams FJ33-5A turbofan

Top speed 350 mph (560 km/h)

The world's first certified single-engine civilian jet was developed by a company that had already challenged and overtaken the leading US light aircraft manufacturers with its innovative, composite construction piston-engine SR20 and SR22.



▽ Airbus A320neo 2016

Origin European consortium

Engine 2 x 27,120 lb (12,300 kg) thrust CFM International LEAP-1A or Pratt & Whitney PW1100G turbofan

Top speed 518 mph (833 km/h)

The narrow-body A320neo (“neo” standing for new engine option) family comprises three types; the A319neo, A320neo, and A321neo. Developed from the old narrow-body A320 family, the neo series offers a 15–20 percent improvement in fuel economy.



▽ Boeing 737 MAX 2017

Origin US

Engine 2 x 29,317 lb (13,298 kg) thrust CFM International LEAP-1B turbofan

Top speed 521 mph (839 km/h)

To compete with the A320neo, Boeing fitted its long-running 737 design with much larger LEAP turbofans. Unfortunately, the change caused instability that Boeing elected to fix with an automatic trim system, resulting in two early crashes with a significant number of fatalities.



No Pilot Required

Rapid advances in battery and electric motor development, computer control and satellite positioning systems have made small autonomous aircraft—drones—a reality. Now the race is on to gain acceptance for much larger eVTOL (electric vertical takeoff and landing) delivery systems and pilotless air taxis.



◁ Volocopter VC200 2016

Origin Germany

Engine 18 x three-phase PM synchronous brushless DC electric motors

Top speed 62 mph (100 km/h)

Now in production as the “optionally piloted” VC 2X, the Volocopter concept is based on a fail-safe, multiple rotor array and complies with European ultralight helicopter rules. A pilotless air taxi derivative, the VoloCity, is under development.